Space Weather-Aware Anomaly Detection for CubeSats

Research Project Brief | SW-ADP

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THE PROBLEM

Traditional CubeSat anomaly detectors analyze only onboard telemetry: voltage, temperature, magnetic field readings. They are blind to the external environment.

When a geomagnetic storm strikes, the detector sees the symptom but not the cause:

- Late detection — anomalies caught after the fact
- High false-alarm rates — operators lose trust in alerts
- No actionable context — operators cannot explain why

WHY NOW

Solar Cycle 25 has already passed its proposed maximum (2024-2025), and only getting more intense. Though predicted to be lighter than the previous solar cycle, it has been the most active period in two decades.

- Increased geomagnetic storm frequency and intensity
- Higher radiation and drag on LEO satellites
- CubeSat anomaly rates demonstrably elevated during storm periods

This is the right moment to build context-aware detection.

PROPOSED APPROACH

Core Idea

Fuse CubeSat telemetry with space weather indices from NOAA (Kp, Dst, F10.7, solar wind) and run anomaly detection on the combined signal. Context-aware detection enables earlier warnings with calibrated confidence scores operators can trust.

Datasets

- OPSSAT-AD (ESA) — labeled anomalies, 9 telemetry channels, established benchmark
- NOAA SWPC — Kp/Dst/F10.7 indices, timestamped, open access
- ESA Anomaly Dataset — cross-mission generalization testing

Methods

- **Baselines:** Isolation Forest + Matrix Profile (STUMPY)
- **Neural:** LSTM Autoencoder - captures temporal dependencies
- **Calibration:** Online conformal prediction on reconstruction error

Core Experiment

Ablation study: every model run with telemetry-only vs. telemetry + space weather features. Measure NAB score, AU-PR, and mean lead-time. If space weather features improve results, the contribution is demonstrated.

WHAT SUCCESS LOOKS LIKE

+10%	30s+	>=90%	Reduced
NAB Score	**Lead Time**	**Coverage**	**False Alerts**
vs. telemetry-only baseline	*before anomaly onset*	*calibrated alert confidence*	*via space weather context*

DELIVERABLES

- Reproducible Python pipeline (open-source, version controlled)
- Ablation benchmark: telemetry-only vs. +space weather
- Operator dashboard — anomaly overlays, confidence, explanations
- Research paper – Polished and publishable
- Research poster — 17th Annual Student Poster Competition (April 2026)
- Live demo — Boeing Engineering Design Expo (May 2026)

BROADER IMPACT

- Fewer surprise safe-modes and less mission loss for CubeSat operators
- Calibrated AI keeps operators in control — the system informs, it does not decide
- Reusable pipeline for Saudi Space Agency and university flight programs
- Reduced space debris risk from uncontrolled satellites during storm events
- Lowers barrier to resilient space operations for emerging space nations

– Stretch goal: TinyML feasibility on ESP32-S3 (on-board inference)

TIMELINE

Mar 4-17	Data pipeline, baseline models, ablation study, preliminary results
Mar 18-29	Figures finalized, poster designed and submitted for faculty review
Apr 2	17th Annual Student Research Poster Competition — in-person judging
Apr - May	LSTM Autoencoder, conformal calibration, full dashboard, demo
May 2026	Boeing Engineering Design Expo — live demonstration